

ENSIA Nexus

AI-Enhanced Research Collaboration Platform

Project Report

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Introduction

Overall Motivation

Higher-education institutions such as ENSIA host a rich and diverse research ecosystem, comprising advanced laboratories, focused research groups, ongoing academic projects, and motivated students eager to participate. However, the mechanisms connecting these entities are often fragmented. Teachers create projects within research groups but lack a centralized way to publicize them. Students must rely on word-of-mouth to discover opportunities, task tracking is scattered across generic tools, and candidate evaluation is manual, subjective, and time-consuming.

ENSIA Nexus bridges this gap through a unified, web-based research collaboration platform designed specifically for academic environments. The platform provides:

- A **single hub** for research labs, groups, projects, tasks, publications, and collaboration calls.
- **AI-powered candidate ranking** to assist teachers in objectively and efficiently evaluating student applications.
- **Real-time communication** and structured task management tailored to academic project hierarchies.

Existing Solutions

Tool / Method	Purpose	Limitations
Email & Word-of-mouth	Project discovery	No structure, high search friction
Google Sheets / Excel	Application tracking	Manual, lacks secure access controls
Trello / Notion	Task management	Isolated from the academic hierarchy
LMS (Moodle, Classroom)	Course management	Not designed for research workflows
GitHub Projects	Code-centric tracking	Significant technical barrier for non-devs

Table 1: Comparison of existing tools and their limitations.

Generic productivity software fails to support the structured hierarchical organization of academic research (Labs → Groups → Projects), nor does it provide AI-assisted evaluation or institution-wide academic role management.

Main Pain Points Addressed

1. **Fragmented Discovery:** The absence of a unified portal to browse active research projects, open collaboration calls, and published lab papers.
2. **Manual Application Review:** Teachers must evaluate resumes, portfolios, and cover letters manually, leading to inconsistent scoring and administrative bottlenecks.

3. **Absence of Role-Aware Controls:** Existing tools lack fine-grained, academic-specific permissions required to distinguish between lab directors, group leaders, project leads, student members, and external reviewers.
4. **Disconnected Communication Channels:** Team announcements, project boards, and notifications reside in isolated tools, separating discussion from context.
5. **Lack of System Observability:** Difficulties in tracking backend reliability, auditing application issues, or monitoring error rates.

These key issues motivate the design of an **AI-enhanced, role-aware, full-stack platform** optimized specifically for academic research ecosystems.

The Proposed Approach

Problem–Solution Alignment

Pain Point	ENSIA Nexus Solution
Fragmented discovery	Unified Landing & Aggregation Service —A curated, searchable directory of research lines, open calls, and scientific publications.
Manual applicant evaluation	AI-Powered Candidate Ranking —Automated resume, skill, and letter scoring using multi-dimensional matching.
Absence of role-aware controls	Academic Permission Guard —A fine-grained, context-aware RBAC system covering global roles, project roles, and group ownership.
Disconnected communication	WebSocket Notification Hub —Real-time in-platform updates and contextual task notifications.
Lack of observability	Observability Middleware —Structured JSON logging, W3C Trace Context headers, and automated error tracking.

Table 2: Problem–Solution alignment matrix.

System Overview

The platform is designed around a clean, layered client–server architecture that separates interactive user interfaces from core business logic and transactional storage.

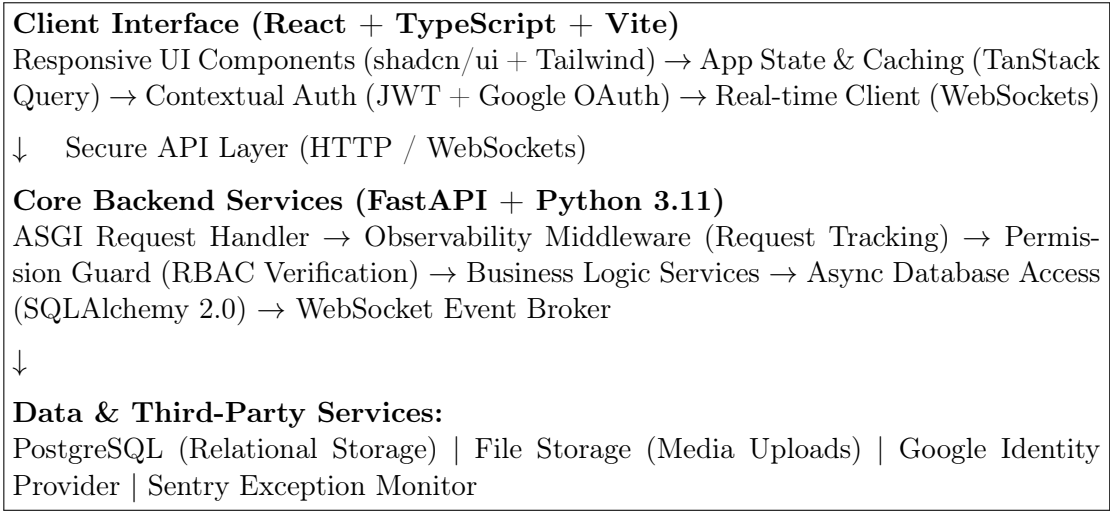


Figure 1: High-level system architecture.

The application data flows securely from user actions on the frontend through to request-scoped tracing, strict permission verification at the service layer, and highly optimized, asynchronous queries to the database. Real-time notifications are instantly pushed back to the client via a lightweight WebSocket broker.

Core Features & Technical Implementation

2.3.1 AI-Powered Candidate Ranking

To streamline recruitment, ENSIA Nexus features an automated scoring engine that evaluates student applications against project criteria. When a student applies, the engine analyzes multiple dimensions to yield a balanced, objective score.

Scoring Formulation:

$$\text{Score}(s) = w_1 S_{\text{cv}} + w_2 S_{\text{skills}} + w_3 S_{\text{projects}} + w_4 S_{\text{motivation}} + w_5 S_{\text{academic}} \quad (1)$$

Where:

- S_{cv} represents CV completeness and formatting profile.
- S_{skills} measures the exact overlap between candidate skills and project prerequisites.
- S_{projects} scores the relevance and scale of the candidate's previous project history.
- $S_{\text{motivation}}$ calculates the similarity between the motivation letter and the project scope (using TF-IDF cosine similarity).
- S_{academic} represents academic standing and institutional metadata.

The weighted variables are normalized to $[0, 1]$, offering teachers a sorted, high-level ranking list with a clear breakdown of each candidate's strengths. Automated constraints prevent spamming by limiting active applications and ensuring proper role assignments upon acceptance.

2.3.2 Unified Technology Stack

The system is built on a modern, robust open-source stack, categorized for modularity and scalability:

Component Layer	Core Technologies	Engineering Rationale
Frontend Web App	React 18, TypeScript, Vite	Native type-safety, rapid compilation, co
UI Styling & Motion	shadcn/ui, Tailwind, Framer Motion	Highly polished, cohesive design system
Backend Framework	FastAPI, Python 3.11	Modern ASGI performance, automatic S
Data Access Layer	PostgreSQL, SQLAlchemy 2.0, asyncpg	Fully asynchronous relational queries, ty
Identity & Security	JWT, Google OAuth, bcrypt	Secure, stateless authentication with ins
System Observability	structlog, W3C Tracing, Sentry	Detailed telemetry, request tracking, and

Table 3: ENSIA Nexus core technology stack and rationale.

2.3.3 Access Control & Permission Guard

A major innovation is the dynamic **Permission Guard**. Rather than using simple static roles, the guard evaluates permissions dynamically based on context. It intercepts incoming API calls, extracts relevant resource identifiers (e.g., a specific Task or Project ID), resolves ownership hierarchies, and verifies permissions dynamically. For instance, if a Teacher attempts to modify a project task, the guard verifies if they are the direct project lead or the overarching Research Group leader, granting access through an elegant, implicit elevation logic.

2.3.4 User Experience Architecture

The frontend features a role-adaptive navigation architecture supporting three distinct user profiles:

- **Public Discovery Portal:** Intuitive, high-performance directory for external visitors and students to search active labs, group portfolios, open collaboration calls, and published papers.
- **Student Workspace Hub:** Interactive dashboards containing a drag-and-drop Kanban project board, resume builders, application trackers, and live notification feeds.
- **Academic Leadership Console:** Management tools for Lab Directors and Teachers to coordinate research lines, manage team rosters, evaluate candidates, and post announcements.

User Interface Showcase

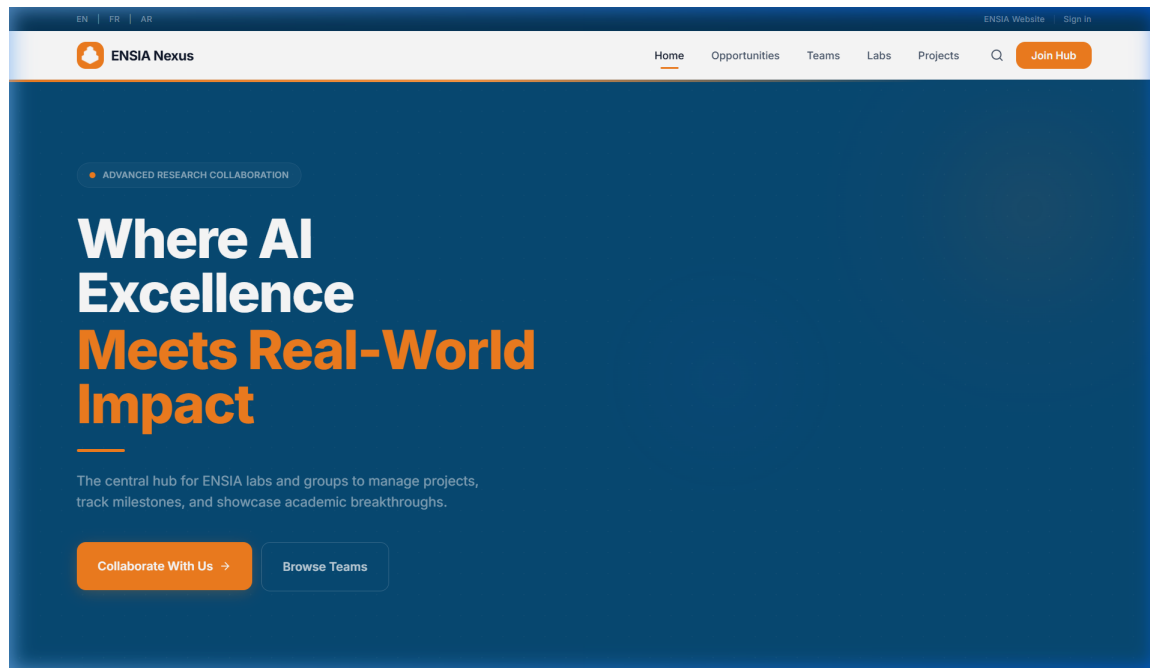


Figure 2: Platform Landing Page—A polished interface showing open research lines, active statistics, and navigation.

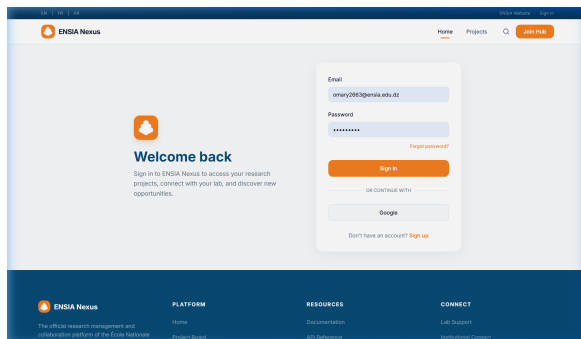


Figure 3: Secure Login interface with support for local credentials and SSO.

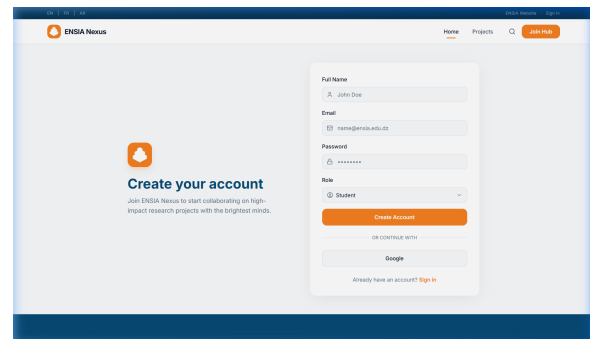


Figure 4: Role selection and onboarding during user registration.

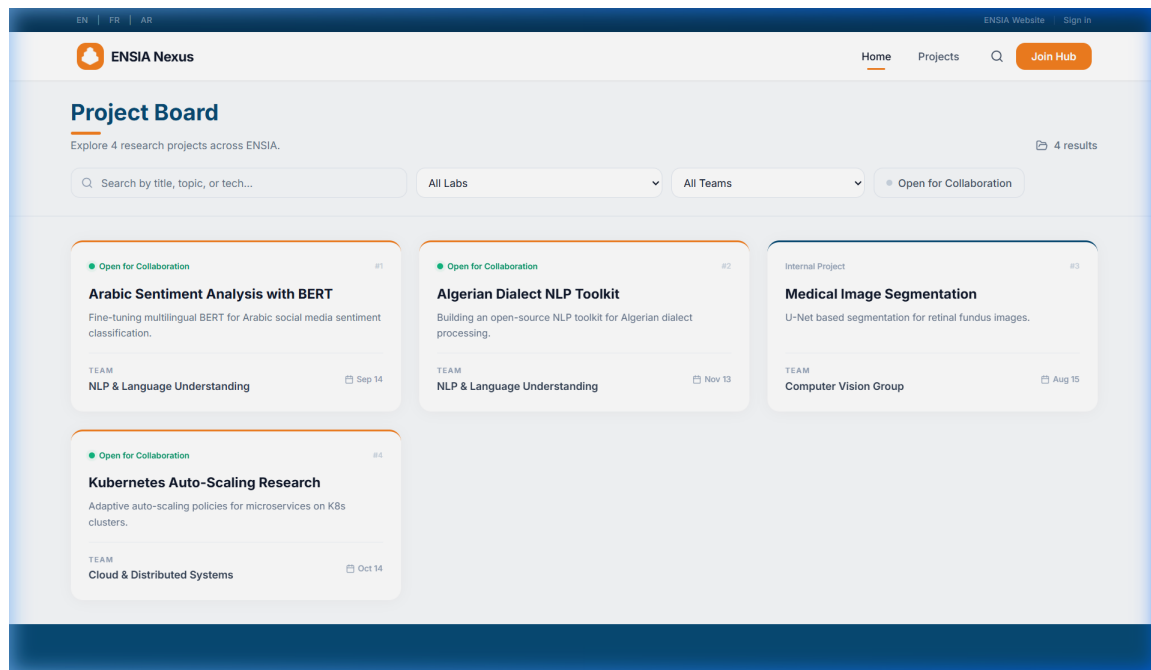


Figure 5: Public Project Catalog—Searchable grid allowing students to filter projects by lab, research field, and application status.

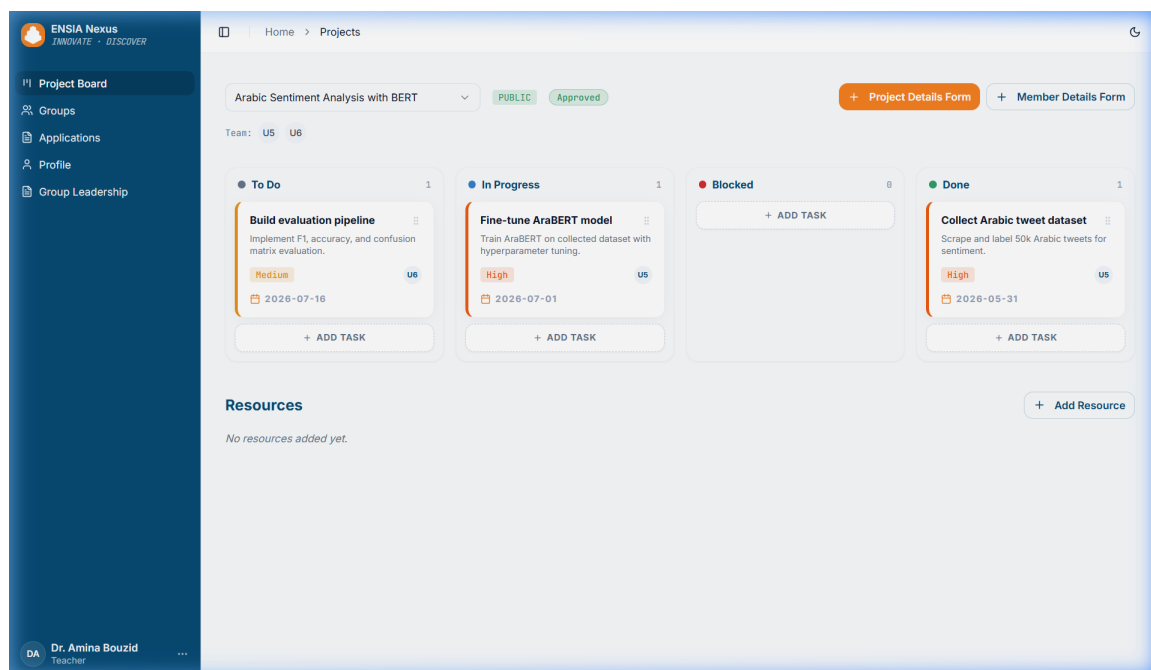


Figure 6: Interactive Project Workspace—A collaborative Kanban board featuring task assignment, priority levels, and progress tracking.

Workspace Management & AI Integration Views:

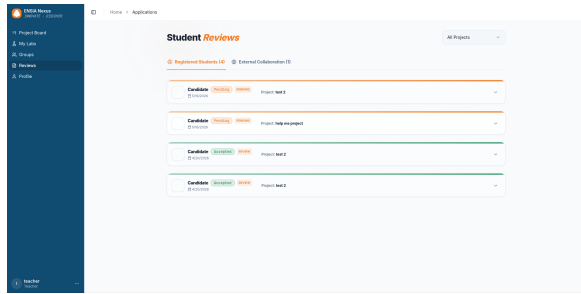


Figure 7: AI Candidate Evaluation—Teacher view showing student applications sorted dynamically by the scoring engine with per-dimension suitability breakdowns.

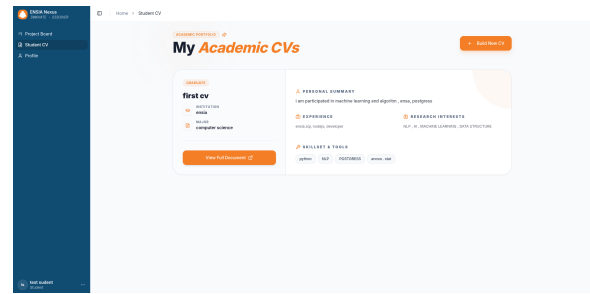


Figure 8: Student CV & Portfolio Builder—A custom interface where students specify skills, past projects, and attach materials to train the matching model.

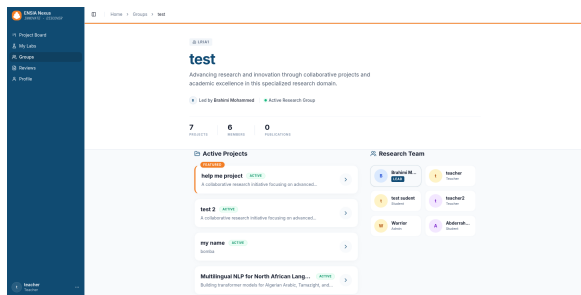


Figure 9: Research Group Console—Group leader interface showing active research projects, calls for collaboration, and member rosters.

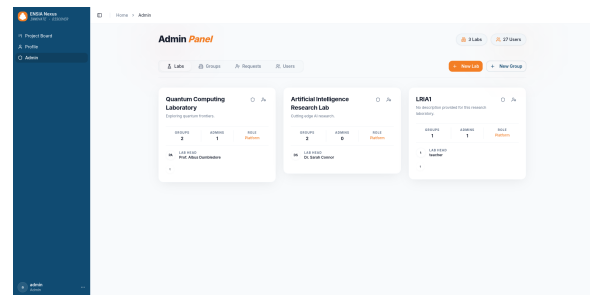


Figure 10: Platform Admin Console—A centralized panel for system metrics, role requests, lab creation pipelines, and security logs.

Verification & Comparative Analysis

Testing and Validation Strategy

The platform’s reliability is backed by a robust suite of over 100 automated test cases. Rather than testing files in isolation, the validation suite verifies critical integration scenarios across three essential architectural pillars:

- **Authorization Scenarios:** Ensuring the Permission Guard successfully isolates concurrent teams. For example, testing that a student from one Research Group receives a strict authorization error (HTTP 403) when attempting to view or modify an unrelated group’s private boards, while a Group Leader gains implicit editor access.
- **Application Rules Concurrency:** Validating business logic constraints under load, such as verifying that duplicate pending application attempts are blocked (HTTP 400), and that the system strictly rejects a student who has reached the maximum threshold of two rejections for a specific project.
- **Input Robustness Edge Cases:** Testing input validation, schema conversion via Pydantic v2, and secure file handling (e.g., verifying that profile picture uploads undergo automatic image validation, size-checking, and resizing while rejecting invalid file formats).

Comparative Analysis

To measure the value of the platform, the proposed system was evaluated against traditional methods currently used in academic environments:

Capability	Traditional Academic Methods	ENSIA Nexus Platform
Project Discovery	Dispersed emails, manual announcements	Centrally aggregated catalog with dynamic search filters
Candidate Review	Subjective review of unstructured emails	Automated AI ranking with clear candidate suitability scoring
Access Control	Simple static permissions (Admin/User)	Context-aware global and project-specific roles
Task Coordination	Generic external tools (Trello, spreadsheets)	Dynamic in-platform Kanban board tied to research groups
Observability	No central application telemetry	Structured JSON logging, W3C trace context, and Sentry tracking
Response Time	Delayed email coordination (days/weeks)	Instant notifications pushed in real-time via WebSockets

Table 4: Comparative analysis: traditional methods vs. ENSIA Nexus.

Project Implementation Scale

To demonstrate the depth and implementation scale of ENSIA Nexus, key structural metrics are summarized below:

- **Architectural Depth:** A rich backend API spanning over 60 fully documented endpoints across 28 distinct functional routers.
- **Data Consistency:** A PostgreSQL database comprising 22 fully normalized relational entities, managed through automated Alembic migrations.
- **Comprehensive Front-End:** A fully realized user interface covering 25 structured views, dynamically rendering based on user roles and project permissions.
- **End-to-End Resolution:** Verification that all main pain points are resolved, ensuring a fast, secure, and observable collaboration workspace.

Conclusion & Future Directions

System Summary

ENSIA Nexus provides a production-grade, full-stack collaborative platform built specifically for higher-education research environments. By successfully organizing academic entities into structured hierarchies (Labs → Groups → Projects) and providing specialized features like AI-powered candidate scoring, contextual RBAC permissions, and real-time WebSocket communication, the platform eliminates the major friction points of traditional methods. It delivers a modern user experience backed by robust, observable backend services.

Engineering Trade-offs Strategic Enhancements

To transition the platform to a highly scalable production environment, key engineering trade-offs have been identified along with their corresponding strategic roadmaps:

Architectural Area	Current Engineering Trade-off	Strategic Roadmap	Production
Caching Layer	Landing page aggregation queries hit the primary PostgreSQL database directly.	Integrate a Redis cache layer to store frequently accessed public views and session state.	
Asset Storage	Uploaded files (resumes, pictures) are stored on the local container volume.	Migrate local storage to an AWS S3 or compatible cloud object storage bucket for persistence.	
Rate Limiting	API does not restrict high-frequency requests.	Implement FastAPI-Limiter middleware to protect public endpoints from API abuse.	
WebSocket Security	Real-time connection handshake runs without initial token validation.	Implement secure JWT verification during the initial WebSocket handshake upgrade.	
AI Scoring Engine	Candidate ranking relies on calibrated static weighting rules.	Transition to a machine learning classifier trained on historical selection decisions.	

Table 5: Current architectural trade-offs and production roadmap.

Appendix & Team Contributions

Team Collaboration & Shared Ownership

The engineering of ENSIA Nexus was carried out collectively by a core development team: **Tarek Sibachir**, **Mohamed Aymen Benmalek**, **Abderrahmen Mehaddi**, and **Guerfi Dhia Eddine**.

Due to the complex, highly integrated nature of the platform, the development followed a highly collaborative, shared-ownership model. Rather than working in isolated silos, all four team members acted as full-stack engineers throughout the entire project lifecycle, co-designing and co-implementing every layer of the platform. Key integrated accomplishments include:

- **Database Architecture & Modeling:** Collaboratively designing the 22 normalized relational PostgreSQL entities, establishing complex foreign key constraints, and managing migrations cleanly via Alembic.
- **Backend Service Engineering:** Jointly developing the FastAPI routing architecture (spanning 60+ endpoints), implementing custom middleware, and crafting the dynamic request-context Permission Guard.
- **Frontend Interface Development:** Co-creating the 25 interactive, role-adaptive views using React, managing client-side state caching with TanStack Query, and styling modern transitions via Tailwind and Framer Motion.
- **AI scoring & Live Telemetry:** Jointly formulating and testing the multi-dimensional candidate suitability equation and building the WebSocket-based server to dispatch live notifications.